Collect data in the order represented below

flag_length = 0.01168 m
u_flag_length = 0.000005 m
u_time = 0.001 s
g = 9.805 m/s^2
u_g = 0.005

Analysis helpers

No data entry needed

Part 1 - Light cart

cart_mass = 0.19146 kg
u_cart_mass = 0.000005 kg

Light cart - hanging-mass setup 1

m_hang = 0.00997 # kg; REPLACE with total hanger + added mass
u_m_hang = 0.000005 # kg; REPLACE with balance uncertainty
v1 = [1.99, 2.02, 2.01, 2.01, 2.00] # m/s, gate 1
t1 = [0.692, 0.810, 0.869, 0.742, 0.778] # s, gate 1 event time
v2 = [3.84, 3.83, 3.83, 3.84, 3.84] # m/s, gate 2
t2 = [1.543, 1.656, 1.715, 1.589, 1.628] # s, gate 2 event time

Light cart - hanging-mass setup 2

m_hang = 0.01275 # kg; REPLACE with total hanger + added mass
u_m_hang = 0.000005 # kg; REPLACE with balance uncertainty
v1 = [2.23, 2.24, 2.20, 2.22, 2.19] # m/s, gate 1
t1 = [0.884, 0.464, 0.516, 0.607, 0.506] # s, gate 1 event time
v2 = [4.29, 4.30, 4.29, 4.29, 4.29] # m/s, gate 2
t2 = [1.643, 1.219, 1.279, 1.369, 1.272] # s, gate 2 event time

Light cart - hanging-mass setup 3

m_hang = 0.00483 # kg; REPLACE with total hanger + added mass
u_m_hang = 0.000005 # kg; REPLACE with balance uncertainty
v1 = [1.39, 1.38, 1.39, 1.39, 1.39] # m/s, gate 1
t1 = [1.426, 1.559, 1.429, 1.298, 1.300] # s, gate 1 event time
v2 = [2.71, 2.70, 2.71, 2.71, 2.69] # m/s, gate 2
t2 = [2.633, 2.773, 2.637, 2.504, 2.512] # s, gate 2 event time

Light cart - hanging-mass setup 4

m_hang = 0.01753 # kg; REPLACE with total hanger + added mass
u_m_hang = 0.000005 # kg; REPLACE with balance uncertainty
v1 = [2.48, 2.61, 2.60, 2.54, 2.53] # m/s, gate 1
t1 = [0.619, 0.801, 0.739, 0.789, 0.810] # s, gate 1 event time
v2 = [4.95, 5.02, 5.02, 4.98, 4.98] # m/s, gate 2
t2 = [1.284, 1.448, 1.386, 1.445, 1.467] # s, gate 2 event time

Light cart - hanging-mass setup 5

m_hang = 0.01945 # kg; REPLACE with total hanger + added mass
u_m_hang = 0.000005 # kg; REPLACE with balance uncertainty
v1 = [2.72, 2.67, 2.70, 2.75, 2.67] # m/s, gate 1
t1 = [0.639, 0.835, 0.672, 0.797, 0.647] # s, gate 1 event time
v2 = [4.98, 5.12, 5.16, 5.27, 5.22] # m/s, gate 2
t2 = [1.261, 1.462, 1.292, 1.413, 1.271] # s, gate 2 event time

Light-cart fit and comparison

No data entry needed

Light-cart checkpoint

- Does the fitted slope agree with the measured cart mass within uncertainty?
  No, the fitted slope does not agree with the measured cart mass within uncertainty.
- Is the intercept consistent with zero?
  Yes, the intercept is consistent with 0.
- Do the individual measured accelerations agree with their predictions?
  Yes, the individual measured accelerations agree with their predictions.
- If not, inspect the track level, pulley, string, gate triggering, and release technique before adding cart mass.